



December 2025

(For CV)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

20 Marks

- 1-

14. Chezy's formula is used for
 A) Laminar flow
 B) Open channel uniform flow
 C) Pipe flow
 D) Compressible flow
15. Most economical rectangular section has
 A) Depth = $B/2$
 B) Depth = B
 C) Width = $2D$
 D) Hydraulic radius = $D/2$
16. Jet impact analysis uses
 A) Bernoulli's equation
 B) Impulse-momentum principle
 C) Continuity only
 D) Hydrostatics
17. Pelton wheel is
 A) Impulse turbine
 B) Reaction turbine
 C) Mixed flow turbine
 D) Axial turbine
18. Kaplan turbine is suitable for
 A) Low head, high discharge
 B) High head, low discharge
 C) Medium head, medium discharge
 D) Very high head
19. Manometer works on principle of
 A) Energy conservation
 B) Hydrostatic pressure
 C) Momentum transfer
 D) Continuity equation
20. Cavitation occurs when
 A) Pressure < Vapour pressure
 B) Pressure > Vapour pressure
 C) Velocity low
 D) Density high

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Derive the expression for differential pressure for a manometer in the following cases:
Case 1: Two pipes at different levels.
Case 2: Two pipes at the same level. | 08 | L*1 | 2 | 1,2 |
| b) A plate 0.025mm distant from a fixed plate, moves at 60 cm/s and requires a force of 2 N per unit area to maintain this speed. Determine the fluid viscosity between the plates. | 08 | L3 | 1 | 2,3 |
| 2. a) Define surface tension and capillarity. Derive expression for capillary rise and capillary fall. | 08 | L2 | 1 | 1,2 |
| b) A rectangular plane surface is 2 m wide and 3 m deep. It lies in vertical plane in water. Determine the total pressure and position of centre of pressure on the plane surface when its upper edge is horizontal and
i. coincides with water surface,
ii. 2.5 m below the free water surface. | 08 | L3 | 2 | 2,3 |
| 3. a) State and prove Pascal's law. | 08 | L2 | 2 | 1,2 |
| b) Calculate the density, specific weight, and weight of one litre of petrol having a specific gravity of 0.7. | 08 | L3 | 1 | 1,2 |

Unit – II

- | | | | | |
|--|----|----|---|-----|
| 4. a) What is mouth piece? with a neat sketch explain classification of mouth pieces. | 08 | L3 | 4 | 1,2 |
| b) An oil of sp. gr. 0.8 is flowing through a venturi meter having inlet diameter 20 cm and throat diameter 10 cm. The oil-mercury differential manometer shows a reading of 25 cm. Calculate the discharge of oil through the horizontal venturi meter. Take $C_d=0.98$. | 08 | L4 | 3 | 2,3 |
| 5. a) Water discharge at the rate of 98.2 lit/sec through a 120mm diameter vertical sharp-edged orifice placed under a constant head of 10m. A point on the jet measured from the vena-contracta of the jet has coordinates 4.5 m horizontal and 0.54 meter vertical. Find the coefficient C_v , C_c and C_d of orifice. | 08 | L3 | 4 | 2,3 |

- | | | | | | |
|-------|---|----|----|---|-----|
| b) | Derive Bernoulli's equation for the flow of an incompressible frictionless fluid from consideration of momentum. | 08 | L2 | 3 | 1,2 |
| 6. a) | Explain water hammer. Derive expression for pressure rise due to sudden closure. | 08 | L3 | 3 | 1,2 |
| b) | Water flows through a triangular right-angled notch first and then over a rectangular notch of 1 m width. The discharge co-efficient of the triangular and rectangular weirs are 0.6 and 0.7 respectively. If the depth of water over the triangular weir is 360 mm, find the depth of water over the rectangular weir. | 08 | L3 | 4 | 1,2 |

Unit – III

- | | | | | | |
|-------|---|----|----|---|-----|
| 7. a) | Derive an expression for the most economical trapezoidal channel sections. | 08 | L3 | 5 | 2,3 |
| b) | A flow of water of 150 liters per second flows down in a rectangular flume of width 600 mm and having adjustable bottom slope. If Chezy's constant C is 55, find the bottom slope necessary for uniform flow with a depth of flow of 300 mm. Also find the conveyance K of the flume. | 08 | L3 | 5 | 1,2 |
| 8. a) | Derive an expression for the force exerted by a jet of water on a fixed inclined plate. | 08 | L3 | 5 | 1,2 |
| b) | A rectangular channel carries water at the rate of 500 litres/s when bed slope is 1 in 2000. Find the most economical dimensions of the channel if C=50. | 08 | L3 | 5 | 2,3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
